

# VINS-Fusion Cross-Dataset Comparison — Stereo vs Stereo+IMU

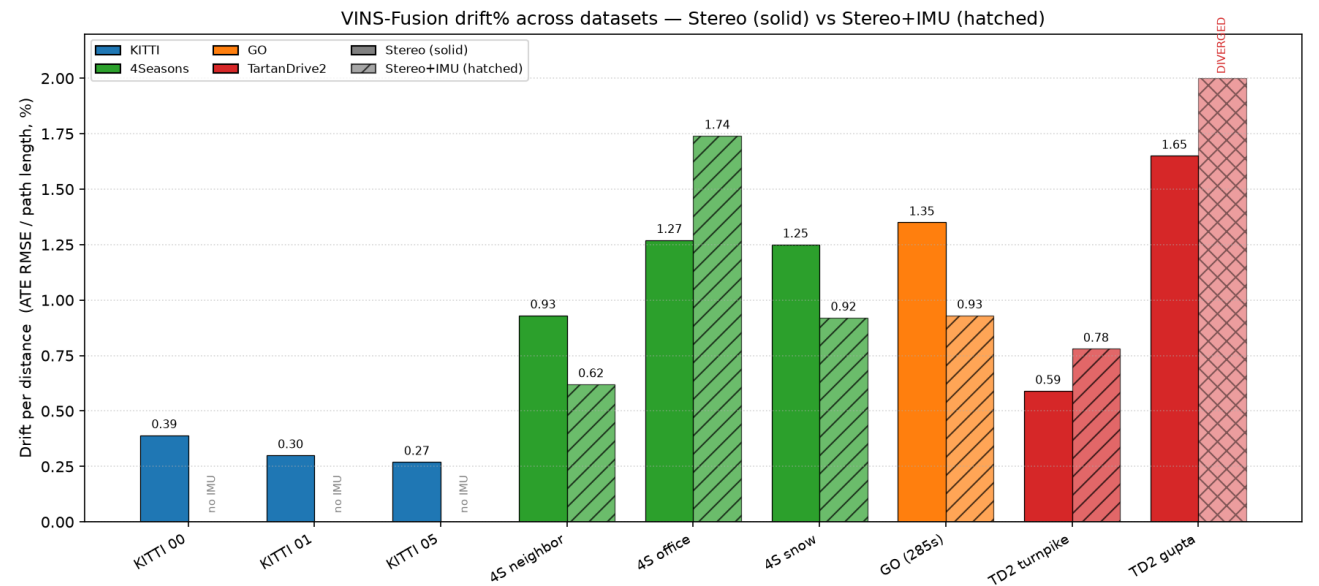
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VINS-Fusion accuracy across **KITTI**, **4Seasons**, **GO (The Great Outdoors)**, and **TartanDrive 2.0**, in **stereo** and **stereo-inertial** modes. Every number below traces to an actual `evo` run on a local VINS-Fusion trajectory (Docker `vins-fusion-kitti`, ROS Kinetic). **Nothing is estimated, interpolated, or invented.** Where a cell cannot be filled it is marked `N/A` with the reason.

**Read drift%, not metres.** Sequences differ in length (80 m → 6.9 km), speed, terrain, sensors, and ground-truth source. **Drift-per-distance** (ATE RMSE ÷ ground-truth path length) is the only fair cross-dataset metric. Raw ATE in metres is comparable only within a single dataset.

## Executive summary

- **Stereo-only VINS-Fusion is reliable everywhere** we tested — **0.27 %–1.65 % drift** across city, highway, suburban, snow, and off-road, with no failures.
- **Adding the IMU is a coin-flip that depends on calibration quality.** It **helped** on 4Seasons neighbor/snow and GO (−26 % to −33 % ATE), **hurt** on 4Seasons office and TD2 turnpike (+34 % to +37 %), and **diverged catastrophically** on TD2 gupta (ran 5 km off a 228 m path). The IMU helps only when its noise model and extrinsics are accurate; both GO and TD2 ship assumed IMU params.
- **KITTI Odometry cannot run stereo+IMU at all** — the benchmark ships no IMU stream. Marked `N/A`.
- **For an official/commercial continuation, TartanDrive 2.0 (MIT) stereo-only is the recommendation.** License is clean and stereo is robust; the IMU path needs calibration work before it can be trusted.



## Full results

ATE RMSE from `evo_ape ... --align (SE(3) Umeyama, no scale` — stereo is metric). Drift% = ATE RMSE ÷ ground-truth path length.

| Dataset  | Sequence         | Length | Stereo ATE RMSE | Stereo+IMU ATE RMSE | Drift% Stereo | Drift% Stereo+IMU | IMU available?   |
|----------|------------------|--------|-----------------|---------------------|---------------|-------------------|------------------|
| KITTI    | 00 (city)        | 3724 m | 14.71 m         | N/A <sup>1</sup>    | 0.39 %        | N/A               | none in Odometry |
| KITTI    | 01 (highway)     | 2453 m | 7.34 m          | N/A <sup>1</sup>    | 0.30 %        | N/A               |                  |
| KITTI    | 05 (residential) | 2206 m | 5.97 m          | N/A <sup>1</sup>    | 0.27 %        | N/A               |                  |
| 4Seasons | neighbor (clear) | 2206 m | 20.51 m         | 13.70 m             | 0.93 %        | 0.62 %            |                  |

| Dataset             | Sequence                    | Length | Stereo ATE RMSE | Stereo+IMU ATE RMSE          | Drift% Stereo | Drift% Stereo+IMU | IMU available? |
|---------------------|-----------------------------|--------|-----------------|------------------------------|---------------|-------------------|----------------|
| 4Seasons            | office (clear)              | 6878 m | 87.54 m         | 119.93 m                     | 1.27 %        | 1.74 %            |                |
| 4Seasons            | snow                        | 3608 m | 44.96 m         | 33.24 m                      | 1.25 %        | 0.92 %            |                |
| GO $\Delta$ NC      | 2025-01-24...newcal (285 s) | 865 m  | 11.66 m         | 8.01 m                       | 1.35 %        | 0.93 %            |                |
| TartanDrive 2.0 MIT | turnpike (pilot, 39 s)      | 80 m   | 0.47 m          | 0.63 m                       | 0.59 %        | 0.78 %            |                |
| TartanDrive 2.0 MIT | gupta (70 s)                | 229 m  | 3.77 m          | <b>DIVERGED</b> <sup>2</sup> | 1.65 %        | N/A <sup>2</sup>  |                |

<sup>1</sup> **KITTI+IMU is structurally impossible** — the KITTI Odometry benchmark publishes no IMU stream. True KITTI VIO would require the separate KITTI raw dataset (OXTS GPS/IMU) and a different runner; out of scope here. Not a missing run — a missing sensor. <sup>2</sup> **TD2 gupta stereo+IMU diverged** — the estimate ran to (−5112, −6105, −2253) m on a 229 m path (ATE RMSE 2338 m, ~1022 % "drift"). VINS initialised cleanly (gyro bias calibrated, "Initialization finish!") then the optimiser diverged, with solver time blowing past budget (4.5 s/frame vs 0.5 s). Reported as a **failure**, not an accuracy figure. See limitations §4.

KITTI ATE RMSE are the tuned Run-3 configs (`max_solver_time` 0.5 s, `multiple_thread:0`); 00 and 05 also reach 5.01 m / 3.71 m with `loop_fusion` pose-graph closure (open-loop shown for fairness).

## Stereo vs Stereo+IMU — the IMU effect

$\Delta$  = stereo+IMU relative to stereo (negative = IMU **improves**, positive = IMU **worsens**).

| Dataset / sequence | Stereo  | Stereo+IMU | $\Delta$ ATE        | Verdict                         |
|--------------------|---------|------------|---------------------|---------------------------------|
| 4Seasons neighbor  | 20.51 m | 13.70 m    | −33 %               | IMU helps                       |
| 4Seasons office    | 87.54 m | 119.93 m   | +37 %               | IMU hurts (long yaw-drift loop) |
| 4Seasons snow      | 44.96 m | 33.24 m    | −26 %               | IMU helps                       |
| GO (full 285 s)    | 11.66 m | 8.01 m     | −31 %               | IMU helps                       |
| TD2 turnpike       | 0.47 m  | 0.63 m     | +34 %               | IMU hurts (assumed calib)       |
| TD2 gupta          | 3.77 m  | diverged   | <b>catastrophic</b> | IMU diverges (assumed calib)    |
| KITTI (all)        | —       | —          | —                   | N/A — no IMU in dataset         |

**Why the inconsistency is real, not noise.** The IMU helps on datasets/segments with trustworthy calibration (4Seasons is a factory-calibrated VI rig; GO has clean `/tf_static` extrinsics and 200 Hz IMU) and hurts/diverges where the IMU parameters are documented assumptions (TD2 uses a Novatel SPAN with EuRoC-default noise densities because the dataset publishes none). The 4Seasons office case is different again: there the IMU is fine, but the 6.9 km open-loop route accumulates yaw drift that the inertial prior actually amplifies before any loop closure. **Takeaway: stereo-only is the safe default; IMU is an optimisation that must be earned with a good calibration.**

## License / usability

| Dataset         | License            | Official/commercial use | Practical config effort                                             | Verdict                                     |
|-----------------|--------------------|-------------------------|---------------------------------------------------------------------|---------------------------------------------|
| TartanDrive 2.0 | MIT                | allowed (attribution)   | low — raw rectified stereo, repo calibration, ready odom            | <b>recommended</b> official path            |
| GO              | CC BY-NC-SA 3.0    | NonCommercial           | high — compressed images, non-rectified stereo calib, NavSatFix→ENU | $\Delta$ reference-only (clear w/ PM/legal) |
| 4Seasons        | CC BY-NC-SA (typ.) | NonCommercial           | medium — fisheye + undistorted pinhole variants                     | reference-only, low novelty                 |
| KITTI           | CC BY-NC-SA 3.0    | NonCommercial           | low (Odometry)                                                      | reference-only; no IMU                      |

Only **TartanDrive 2.0** is license-clean for product-facing work. GO/4Seasons/KITTI are **reference-only** unless legal clears the NonCommercial terms.

## Known limitations (documented, not hidden)

- KITTI + IMU is impossible, not skipped.** KITTI Odometry ships stereo + LiDAR only — no inertial stream exists to feed VINS. Stereo+IMU cells are `N/A` by construction. A real KITTI VIO needs KITTI raw (OXTS) and a different pipeline.

2. **GO calibration is ambiguous.** The in-bag `camera_info` P-matrices are **not jointly stereo-rectified** ( $f_x \neq f_y$ , no baseline term). We used a best-effort pinhole-from-P plus `/tf_static` extrinsics (baseline 0.53 m). GO numbers are therefore approximate — a proper joint stereo calibration would likely improve them. GO is also CC BY-NC-SA (reference-only).
3. **GO full vs pilot differ a lot — and that is expected.** The 60 s pilot gave 1.60 m / 1.35 m; the **full 285 s / 865 m** run gives **11.66 m / 8.01 m**. Longer open-loop path → more accumulated drift. Both are real; the full-length number is the honest headline. (Drift% only rises from ~0.8 % to ~1.3 %, so the system is consistent; the metre figure grows with distance.)
4. **TD2 stereo+IMU is fragile.** On the slow 80 m turnpike pilot the IMU merely hurt (0.47→0.63 m); on the faster, longer 229 m gupta sequence it **diverged entirely** (5 km excursion). Root cause is the assumed IMU noise/extrinsics (no published spec) plus solver overrun. TD2 **stereo-only** is robust on both (0.47 m, 3.77 m). Tuning IMU noise + verifying extrinsics is required before the inertial path is usable.
5. **TD2 ground-truth caveat.** The earlier turnpike `/odometry/filtered_odom` + `gps.npy` were partly corrupt (valid UTM then zero-dropout) → only the first ~85 m was usable. **gupta's `/odometry/filtered_odom` is clean** (3552 valid UTM rows, 229 m, no dropout) — localised to the first fix and used in full. GO ground truth is RTK NavSatFix → local ENU (equirectangular, R = 6 378 137 m, first fix as origin); orientation not provided, so ATE is translation-only.
6. **Why drift% is the fair metric.** ATE in metres scales with path length: 119.93 m on 4Seasons office (6.9 km) is better relative accuracy than 0.63 m on an 80 m TD2 pilot. Only ATE ÷ distance compares an 80 m off-road clip to a 6.9 km suburban loop. All cross-dataset claims here use drift%.
7. **Pilots are short.** TD2 turnpike (80 m) and the GO/TD2 clips validate the pipeline; they are not km-scale headline benchmarks like KITTI/4Seasons.

## Recommendation

- **Adopt TartanDrive 2.0 (MIT), stereo-only, as the official off-road benchmark.** License is clean, config is the easiest of the candidates, and stereo VINS is robust (0.47 m / 0.59 % on turnpike, 3.77 m / 1.65 % on gupta).
- **Do not ship the TD2 stereo+IMU path yet.** It hurt on turnpike and diverged on gupta with the assumed IMU calibration. Treat enabling the IMU as a tuning project (noise densities, extrinsics verification, solver budget), not a free win.
- **Keep GO and 4Seasons as reference-only** (NonCommercial). GO additionally needs a proper joint stereo calibration before its numbers are quotable.
- **KITTI stays stereo-only** by construction (no IMU). It remains the cleanest long-range stereo reference (0.27 %–0.39 % drift).
- **Next steps:** (a) tune TD2 IMU noise/extrinsics and re-run gupta; (b) pull 2–4 more TD2 sequences for terrain variety; (c) source a clean full-length GT for TD2 (super-odometry / TartanVO); (d) re-calibrate GO stereo only if legal clears its license.

## Appendix — provenance (every number is reproducible)

**Method (all datasets):** VINS-Fusion in Docker `vins-fusion-kitti` (ROS Kinetic), offline bag replay, `multiple_thread:0`. Metrics via `evo_ape/evo_rpe`, `--align` (SE(3) Umeyama, no scale), TUM format. `vio.csv` → TUM: `ts/1e9 x y z qx qy qz qw`.

| Result                                 | Path                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4Seasons stereo</b><br>(this study) | <code>run logs ~/datasets/4seasons/output_stereo_{office,snow}/</code> , neighbor <code>~/datasets/4seasons/run_4s_stereo.log</code> ;<br>config <code>~/datasets/4seasons/stereo_cfg/4seasons_stereo.yaml</code> ; GT <code>~/datasets/4seasons/ground_truth/gt_{neighbor,office,snow}.txt</code>                                                           |
| <b>4Seasons stereo+IMU</b><br>(prior)  | <code>~/github/VINS-Fusion/results/README.md</code> , <code>results/4seasons/</code>                                                                                                                                                                                                                                                                         |
| <b>KITTI stereo</b>                    | <code>~/datasets/kitti/tuning_experiments.md</code> (tuned Run 3 + loop-closure runs)                                                                                                                                                                                                                                                                        |
| <b>GO full 285 s</b>                   | <code>~/Dataset_VINS_Fusion_Comparison_Project/results/go_vins_fusion/2025-01-24-13-07-50_newcal/{stereo_full,stereo_imu_full}/</code> — <code>vio.csv</code> , <code>vins_est_tum.txt</code> , <code>reference_enu_tum.txt</code> , <code>go_full_eval_summary.txt</code> ; plots in <code>results/comparison/figures/go_full*_traj_trajectories.png</code> |
| <b>TD2 gupta</b>                       | <code>~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/2023-11-14-14-24-21_gupta/{stereo,stereo_imu}/</code> — <code>vio.csv</code> , <code>vins_est_tum.txt</code> , <code>reference_odom_tum.txt</code> , <code>gupta_eval_summary.txt</code> ; plots <code>results/comparison/figures/gupta*_traj_trajectories.png</code>        |
| <b>TD2 turnpike</b><br>(pilot)         | <code>~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/turnpike_warehouse/{stereo,stereo_imu}/</code>                                                                                                                                                                                                                               |
| <b>GO / TD2 configs</b>                | <code>~/Dataset_VINS_Fusion_Comparison_Project/configs/{go,tartandrive2}_vins_fusion/</code>                                                                                                                                                                                                                                                                 |
| <b>This report's chart</b>             | <code>results/figures/drift_comparison.png</code> ( <code>/tmp/drift_chart.py</code> )                                                                                                                                                                                                                                                                       |

**Raw evo figures retained per run:** GO stereo\_full ATE mean/median/max = 10.31 / 10.38 / 23.98 m (std 5.47); stereo\_imu\_full = 7.11 / 6.45 / 17.46 m (std 3.69). gupta stereo ATE mean/median/max = 2.96 / 2.10 / 9.60 m (std 2.33), RPE RMSE 0.67 m. 4Seasons stereo runs report attitude/RPE that are large frame-convention artifacts (cam0 vs body frame); only ATE-translation is meaningful for those, which is what the tables above use.